January 2026 CSE 314

Online Assignment on Bash Scripting

Time: 30 minutes

Subsection B2

You are organizing files for the World Cup final broadcast package. The files directly inside `input_dir` must be separated according to their sizes so that the production team can choose an appropriate transfer method.

Command

```
./online-2205XXX.sh input_dir output_dir
```

Output

1. Create the following directories:

   ```
   output_dir/
   ├── Small/
   ├── Medium/
   └── Large/
   ```

2. Consider only non-empty regular files directly inside `input_dir`. Do not search subdirectories.
3. Calculate each file's size in bytes.
4. Copy each file according to its size:
   - `Small/` if its size is less than 50 bytes
   - `Medium/` if its size is from 50 through 99 bytes
   - `Large/` if its size is 100 bytes or more
5. Keep every copied file's original filename.
6. Create `output_dir/sizes.txt`. Add one line for every copied file:

   ```
   size_in_bytes:original_filename
   ```

   Sort the lines numerically by size in descending order.

Sample Input (`input_dir/`)

```
input_dir/
├── lineup.txt       (32 bytes)
├── tactics.dat      (76 bytes)
├── full_report.log  (138 bytes)
├── schedule.txt     (47 bytes)
└── empty.tmp        (empty — IGNORE)
```

Sample Output (`output_dir/`)

```
output_dir/
├── Small/
│   ├── lineup.txt
│   └── schedule.txt
├── Medium/
│   └── tactics.dat
├── Large/
│   └── full_report.log
└── sizes.txt
```

Contents of `sizes.txt`:

```
138:full_report.log
76:tactics.dat
47:schedule.txt
32:lineup.txt
```

Assumptions

- All files directly inside `input_dir` have unique filenames.
- No two processed files have the same size.